

Course: Linear Algebra
Date: 19-11-25

V. Gali
(8)

Quiz: III

Total Time (Minutes): 8

Total Marks: 8

Course Instructor
Ayesha Hafeez Sheikh

Name: _____ Roll No: _____ Section: _____

✓ If T is The transformation from $\mathbb{R}^n \rightarrow \mathbb{R}^m$
i.e. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$, Then for $m = n$
 T is called an linear operator on $\mathbb{R}^n$.

• If A and B are similar invertible matrices,
Then A^{-1} and B^{-1} are similar.

✓ must be true . must be wrong

• may or may not be true . None

✓ If $S = \{v_1, v_2, \dots, v_n\}$ is an orthogonal set of
non zero vectors in an inner product space,
Then S is linearly independent.

• If $U = \begin{bmatrix} 3 & -2 \\ 4 & 8 \end{bmatrix}$ & $V = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$,

Then The standard inner product

= 7